

App Development Project Report

AppDev2 Project Report

1. Student Details

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About Me: I am in the last term of my diploma level(hopefully) and pursuing this degree and my standalone degree

2. Project Details

Project Title: Placement Portal

Problem Statement:

A Placement Portal Application (PPA) web application that allows Admin (Institute), Companies, and Students to interact with the system based on their roles.

Approach:

A Single-Page Application (SPA) built with Vue 3 (frontend) and Flask (backend). Role-based access control separates the Student, Company, and Admin workflows. Background tasks via Celery and Redis handle scheduled jobs such as daily interview reminder emails and monthly placement report generation.

3. AI/LLM Declaration

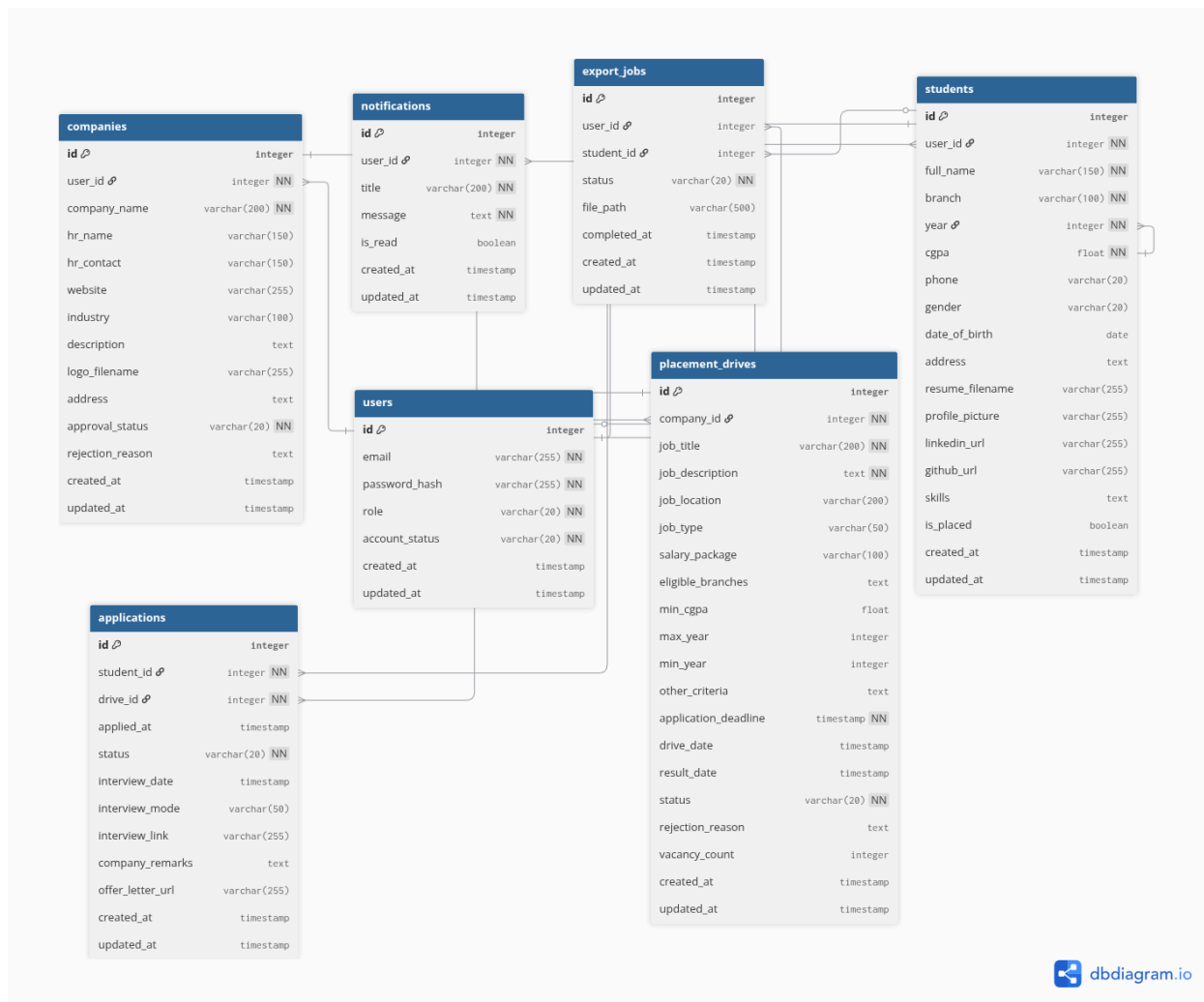
I used GitHub Copilot to assist with css and design and Vue component structure. The extent of AI/LLM usage is around 15–20%, limited to code suggestions and repetitive patterns. All business logic, routing design, and integration were implemented manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework

Flask SQLAlchemy	Object Relational Mapper for SQLite database
Flask-JWT-Extended	JWT-based authentication & role-based authorization.
Flask-Mail	Email notifications (interview reminders, monthly reports)
Flask-Caching Redis	Server-side API response caching
Celery + Redis	Async background tasks & scheduled job queue
Vue 3	Frontend SPA framework
Vue Router	Client-side routing
Pinia	Frontend state management
Axios	HTTP client for REST API calls
Vite	Frontend build tool

5. Database Schema / ER Diagram



6. API Resource Endpoints

Have added the api.yaml file.

7. Architecture and Features (optional)

- server/main.py – Flask application entry point
- server/src/models.py – SQLAlchemy database models
- server/src/apis/ – Flask Blueprints (auth, student, company, admin)
- server/src/jobs/ – Celery app and scheduled/async tasks
- server/src/helpers/ – Auth guards, caching, email, utilities
- client/src/views/ – Vue page components per role
- client/src/stores/ – Pinia state stores (auth, student, company, admin)
- client/src/router/ – Vue Router with role-based navigation guards

Implemented Features:

- Role-based JWT authentication (Student / Company / Admin)
- Student registration, profile editing, and resume upload
- Company registration with admin approval workflow
- Placement drive creation (company) with admin approval
- Eligibility checks (branch, CGPA, year) before applying
- Application status lifecycle: Applied → Shortlisted → Selected / Rejected
- In-app notifications for application status updates
- Admin dashboard with aggregate statistics
- Async CSV export of applications via Celery
- Daily interview reminder emails (Celery Beat)
- Monthly placement report generation and email dispatch
- Redis-based API caching

8. Video Presentation

Drive Link:

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